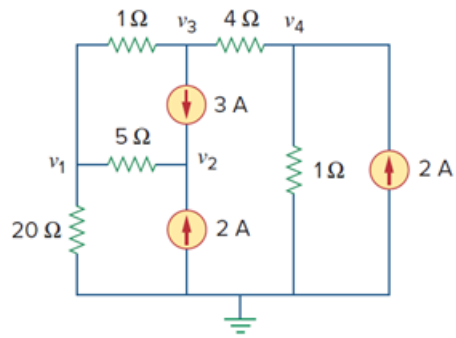


Problem

By inspection, obtain the node-voltage equations for the circuit in Fig. 3.28.

Figure 3.28:



Step-by-step solution

Step 1 of 4

Draw the circuit diagram.

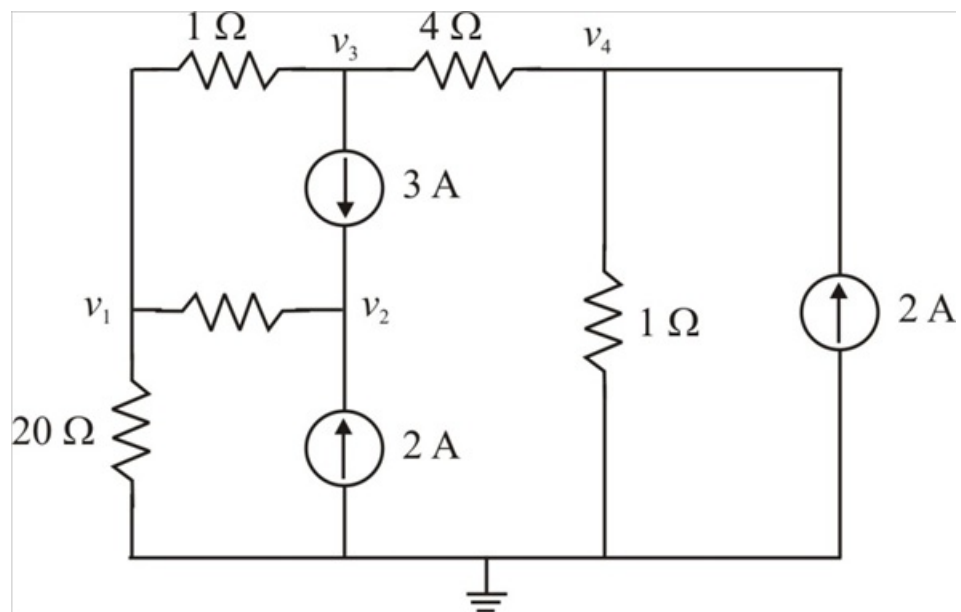


Figure 1

Step 2 of 4

The circuit in Figure 1 has four non reference nodes, so we need four node equations. This implies that the size of the conductance matrix $\mathbf{G}$, is 4 by 4. The diagonal terms of $\mathbf{G}$, in Siemens, are:

$$G_{11} = \frac{1}{20} + \frac{1}{5} + \frac{1}{1} \\ = 1.25$$

$$G_{22} = \frac{1}{5} \\ = 0.2$$

$$G_{33} = \frac{1}{1} + \frac{1}{4} \\ = 1.25$$

$$G_{44} = \frac{1}{1} + \frac{1}{4} \\ = 1.25$$

Step 3 of 4

Determine the off-diagonal terms in the first row.

$$G_{12} = -\frac{1}{5} \\ = -0.2$$

$$G_{13} = -\frac{1}{1} \\ = -1$$

$$G_{14} = 0$$

Determine the off-diagonal terms in the second row.

$$G_{21} = -\frac{1}{5} \\ = -0.2$$

$$G_{23} = 0$$

$$G_{24} = 0$$

Step 4 of 4

Determine the off-diagonal terms in the third row.

$$\begin{aligned}G_{31} &= -\frac{1}{1} \\ &= -1\end{aligned}$$

$$G_{32} = 0$$

$$\begin{aligned}G_{34} &= -\frac{1}{4} \\ &= -0.25\end{aligned}$$

Determine the off-diagonal terms in the fourth row.

$$G_{41} = 0$$

$$G_{42} = 0$$

$$\begin{aligned}G_{43} &= -\frac{1}{4} \\ &= -0.25\end{aligned}$$

The input current vector **I** has the following terms, in amperes:

$$i_1 = 0$$

$$\begin{aligned}i_2 &= 2 + 3 \\ &= 5\end{aligned}$$

$$i_3 = -3$$

$$i_4 = 2$$

Therefore, the node-voltage equations are

$$\begin{bmatrix} 1.25 & -0.2 & -1 & 0 \\ -0.2 & 0.2 & 0 & 0 \\ -1 & 0 & 1.25 & -0.25 \\ 0 & 0 & -0.25 & 1.25 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ -3 \\ 2 \end{bmatrix}.$$